DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND WASHINGTON
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J&A No. N40080-25-J&A-009

JUSTIFICATION AND APPROVAL FOR USE OF OTHER THAN FULL AND OPEN COMPETITION

1. Contracting Activity

Naval Facilities Engineering Systems Command (NAVFACSYSCOM), Washington

2. Description of the Action Being Approved.

This Justification and Approval (J&A) authorizes and approves the use of brand name Safespill Ignitable Liquid Drainage Floor Assemblies (ILDFA) Fire Suppression System within the P-691 Aircraft Development and Maintenance Facility, a Design-Bid-Build project at Naval Air Station Patuxent River, MD.

3. Description of Supplies/Services.

The Government is seeking to provide a Safespill ILDFA Fire Suppression System throughout the hangar bay areas for project P-691. Safespill's proprietary system uses a passive containment system of hollowed, aluminum extrusion profiles with a perforated top to contain and flush away any flammable liquid spill or fire with water, instead of gases or chemicals. The system uses the existing slope of the hangar to drain spilled liquid through channels into covered trenches. Those trenches slope outward into a drainage box which is then extracted and pumped into an exterior containment tank. The existing slope of a typical hangar is about 1% which makes it a relatively slow drain rate, so the ILDFA system uses a water-based flushing system to accelerate the drainage rate.

The estimated amount for the Safespill ILDFA Fire Suppression System is approximately [redacted]. This amount is a nominal percentage, approximately [redacted], of the overall cost for the project (currently estimated [redacted] The type of funds used for the entire project are Fiscal Year 2025 Military Construction funds.

Description	Cost/Unit	Qty	Total
Safespill ILDFA	[redacted]	1	[redacted]

4. Statutory Authority Permitting Other Than Full and Open Competition.

The statutory authority permitting other than full and open competition is Title 10, U.S.C. 3204(a)(1), as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, "Only one responsible source and no other supplies or services will satisfy agency requirements."

5. Rationale Justifying Use of Cited Statutory Authority

Historically, many aircraft hangars have relied on Aqueous Film Forming Foam (AFFF) fire suppression systems due to their effectiveness in quickly extinguishing Class B (flammable liquid) fires. However, the prevalence of AFFF systems is changing due to environmental and public health concerns. Per- and polyfluoroalkyl substances (PFAS) chemicals are known to be found in AFFF and when these fire suppression systems go off, the AFFF runs into sewer systems which consequently can pollute local drinking water.

In response to this growing public health concern, the National Defense Authorization Act for Fiscal Year 2020 (NDAA) was instated to prohibit the purchase of AFFF concentrate after 1 October 2023 and prohibit its use after 1 October 2024. Current Interim Technical Guidance (ITG) FY23-02 “Navy and Marine Corps Facilities with Aqueous Film Forming Foam (AFFF) Fire Suppression Systems” and CNICNOTE 11320 “Transition From Aqueous Film Forming Foam Use Ashore” call for the use of ILDFA in new construction projects.

ILDFA is a technology that is an alternative to Aqueous Film Forming Foam (AFFF) Fire Suppression Systems. It is an environmentally safe alternative to AFFF because it utilizes water only. Flushed water and ignitable liquids are directed to the oil/water separator system that processes a hangar’s normal operational liquids. The system maintains its environmental compliance by utilizing water over potentially harmful chemicals and operates in lower sustainment costs over time as follows:

a. This water-based system is a sustainable, cost-effective, and safe alternative to chemical-based suppression systems, particularly in light of increasing environmental regulations. Unlike AFFF, which requires costly containment and disposal of contaminated runoff, the Safespill ILDFA system produces fewer hazardous byproducts, minimizing long-term expenses. Water does not leave significant residue which simplifies post-activation cleanup and reducing the risk of equipment corrosion or damage.

b. A proprietary function of the Safespill ILDFA system is the zoned flushing which activates flushing only in the zone in which a spill has occurred. The flushing system operates at a quarter gallon per minute or one liter per minute into each channel. Each floor zone needs a water supply of 50 gallons per minute or 200 liters per minute. The maximum water supply, if all zones were employed, would be 200 gallons per minute or 760 liters per minute. The hangar could supply water in this capacity from a standard, domestic water connection. This requires far less water infrastructure and less maintenance.

The National Fire Protection Association (NFPA) 409 Standard on Aircraft Hangars requires low-level systems for aircraft protection. It contains a list of fire suppression systems, including different types of foam systems and Ignitable Liquid Drainage Floor Assemblies (ILDFAs) as acceptable systems. However, there is no MILSPEC foam listed on the Department of Defense (DoD) Qualified Products Database (QPD) under MIL-PRF-24385 for fixed suppression systems

that are currently acceptable to the Navy. The MILSPEC Synthetic Fluorine-Free Foam (SFFF) products listed on the QPD are for fire truck applications for firefighting use.
To utilize MILSPEC SFFF for fixed systems in building applications, it must pass UL162 testing and be listed by Underwriters Laboratories (UL). The UL is a global safety certification organization that tests and certifies products to ensure they meet specific safety, performance, and regulatory standards. There are currently no MILSPEC SFFF products that are UL162 listed.

As a result, the ITG FY23-02 requirements described within each option of (6)(3)(a.-b.) govern the design specifications to plan for ILDFA installation in P-691 in lieu of any other system alternatives to AFFF. The ITG recognizes only one Navy-compliant alternative to AFFF, and that SFFF replacement is an interim solution only. While there are other current alternatives to AFFF on the market, such as water mist systems, they are not yet compliant with DoD fire code for hangars with fueled aircraft.

P-691 requires ILDFA from Safespill because it is the only system available that meets its need for, and achieves the benefits of, environmental safety and compliance with ITG FY23-02. There are no known, commercially available, certified vendors capable of providing ILDFA fire suppression flooring. Additionally, installing SFFF at this stage will lead to increased costs and extended timelines, as it will eventually need to be replaced with ILDFA. Prolonging procurement to entertain non-compliant alternative fire suppression systems for P-691will put aircraft storage, maintenance, and other critical mission operations at risk.

<u>6. Description of Efforts Made to Solicit Offers from as Many Offerors as Practicable.</u>

Per FAR 5.201, a Sources Sought announcement was synopsized on the Sam.gov website on 14 January 2025 to determine if there are other manufacturers capable of providing ILDFA fire suppression system for P-691 flooring. The Sources Sought Notice closed on 22 January 2025. There were no responses received from industry. By receiving no other responses from competing vendors, it indicates that there are no other sources capable of meeting the Government’s requirement to install ILDFA in P-691.

<u>7.Determination of Fair and Reasonable Cost.</u>

Based on the information provided within this J&A, the Contracting Officer has determined the anticipated cost to the Government of the supplies covered by this J&A to be fair and reasonable.

<u>8.Actions to Remove Barriers to Future Competition</u>

The Government will continue to monitor industry to identify other available sources capable of the manufacture and installation of fire suppression system equipment that complies with current and applicable fire suppression system requirements.

CERTIFICATIONS AND APPROVAL

TECHNICAL/REQUIREMENTS CERTIFICATION

I certify that the facts and representations under my cognizance that are included in this Justification and its supporting acquisition planning documents, except as noted herein, are complete and accurate to the best of my knowledge and belief.

Technical Cognizance:

[redacted]

LEGAL SUFFICIENCY REVIEW

I have determined this Justification is legally sufficient.

[redacted]

CONTRACTING OFFICER CERTIFICATION

I certify that this Justification is accurate and complete to the best of my knowledge and belief.

[redacted]

ECH IV COMMANDING OFFICER CERTIFICATION

Upon the basis of the above justification, I hereby approve, as Chief of Contracting, the solicitation of the proposed procurement(s) described herein using other than full and open competition, pursuant to the authority of 10 U.S.C. 2304(c)(1).

[redacted]